..: Map Of CS Department :..

Problem Formulation:

Component of the Problem:

Initial State:-

Lab 02

Goal State:-

Room 03

State Space:-

Labs; Rooms

Action:-

Move Lab 02 to Room 03

Transition State:-

Connected Rooms and Labs

Path Cost:-

Cover distance.

-: Lab 02 TO Room 03 :-

Path From Initial to Goal state:-

Initialization:-
Q: [lab 02]
V: [lab 02]

Dequeue lab 02 & Expand:
Q: [Main Gate, Hall, Lab 01]
V: [lab 02, Main gate, Hall, Lab 01]

Dequeue Main Gate & Expand:
Q: [Hall, Lab 01, Teacher room 02]
V: [lab 02, Main gate, Hall, lab 01, Teacher room 02]

Dequeue Hall & Expand:
Q: [lab 01, Teacher Room 02, Room 03]
V: [lab 02, Main gate, Hall, Lab 01, Teacher Room 02, Room 03].

Dequeue Lab 01 & Expand:
Q: [Teacher Room 02, Room 03, office, Room 02].
V: [Lab 02, Main gate, Hall, lab 01, Teacher Room 02, office].

Dequeue Teacher Room 02 & Expand:
Q: [Room 03, office, Room 02, Teacher Room 01]
V: [Lab 02, Main gate, Hall, Lab 01, Teacher Room 02, Room 03, office, Room 02, Teacher Room 01].

Dequeue office & Expand:
Q: [Room 02, Teacher Room 01, Room 01, Room 03].
V: [Lab 02, Main Gate, Hall, Lab 01, Teacher Room 02, office, Room 02, Teacher Room 01, Room 01, Room 03]

Dequeue R2 and Expand:-
Q: [Teacher Room 01, Room 01, Room 03]
V: [Lab 02, Main gate, Hall, Lab 01, Teacher Room 02, office, Room 02, Teacher Room 01, Room 01, Room 03].

Dequeue Teacher Room 01 & Expand:
Q: [Room 01, Room 03]
V: [Lab 02, Main gate, Hall, Lab 01, Teacher Room 02, office, Room 02, Teacher Room 01, Room 01, Room 03].

Dequeue Room 01 & Expand:

Q: [R0]

V: [Lab 02 , Main gate , Hall , Lab 01 ,
Teacher Room 02 , Office , Room 02 ,
Teacher Room 01 , Room 01 , Room 03].

Shortest Path of Lab 02 to Room 03:

Lab 02 -> Office -> Room 03